

Bond Pricing 101

As a bond provides a contractual right to a series of future payments received at specified points of time, the price for a bond is simply the present discounted value of the future cash flows. The face value of a bond will be repaid at maturity.

A *zero-coupon bond* provides only a bond's face value, and it will be sold at a discount to the face value in order to provide a return and compensate for the risks related to holding it. A *coupon bond* provides the face value at maturity in addition to a series of coupon payments (often on a semiannual basis) until the maturity date. The *coupon rate* is contractually defined as a percentage of the face value.

The *yield to maturity* is the internal rate of return an investor will earn by holding a bond to maturity and receiving its cash flows. The yield to maturity for a new investor differs from the coupon rate whenever the bond sells for a different price than its face value.

Exhibit 7.1 provides a simple example to understand the pricing process for bonds. The bond being considered is a ten-year coupon bond with a face value of \$1,000 and a coupon rate of 2 percent. In this simple example, one coupon payment of \$20 (2 percent of \$1,000) is made at the end of each year for ten years to the bond's owner(s) on those dates, and the face value is paid in full at the end of the tenth year. These can be seen in the exhibit's "Cash Flows" column. The next three columns provide the discounted value of these cash flows for different interest rates: 2 percent, 2.5 percent, and 3 percent. When the discount rate is 2 percent, we see that the total discounted cash flows add to \$1,000, which is the same as the face value. This is an important point: When the interest rate is the same as the bond coupon rate, the price of a coupon bond will match its face value.

Let's be clear about what the discounted value of the payments means. In year ten, for instance, the discounted value of the payment with a 2 percent interest rate is \$836.76. Imagine placing this amount in a bank account that earns an annually compounded 2 percent return each year. After ten years, it